

21 – M1 Party, Tenancy & Security

Module design for the module everything scopes on: legal entities and their trading relationships, locations and till positions, devices, users, the permission catalogue and role builder, external time-boxed access, and the visibility policy the database enforces.

Field	Value
Document	21 – M1 Party, Tenancy & Security (Stage 2 module design; template of doc 20 section 5)
Version	v0.1 – draft for design review; contract freeze targeted end of Stage 2 week 2
Date	8 September 2026
Owning developer	Module developer (full-stack): backend and web screens end to end; the till track consumes the permission snapshot only
Inputs	05 SRS (FR-PTY, FR-SEC, FR-GOV); 06 NFR (security, compliance); 07 Matrices (roles, RBAC, configuration register); 16 answers (A, M, N sections); 09 ADR-02, 04, 12, 18, 26; 17 v0.2.1; 18 part A and section 3.7; 19 sections 1–3; 20 M1 blueprint; 30 section 5.1; 32 sections 2 and 8
Depends on	Cross-cutting services only (identity provider, audit facade, event backbone, configuration register)
Depended on by	Every other module: scope resolution, permission resolution, relationship lookup, location and device facts

Why this module is first and why it is small. Every row in the system carries an owning entity; every request carries a scope and a permission set; every trading document names a relationship. M1 is where those facts are created and nowhere else. It has almost no business process of its own, which is exactly why it must be frozen first: a change here ripples through all nine modules.

1. Purpose and scope

M1 answers four questions for the rest of the system: who exists (entities, locations, till positions, devices), who trades with whom on what terms (relationships), who may do what (users, roles, permissions, grants), and what each caller may see (scope and policy class). It owns the reference data and the policy; the kernel enforces the policy on every request and database session (doc 19 sections 1 and 3, doc 18 section 3.7).

In scope (phase 1)	Out of scope
Entity register (Federation, distributors, MPCs) with tier, VAT registration, district, financial-year settings, costing policy, default language, status	Cooperative membership, shares, governance records (K-01 to K-07)
Trading relationships with effective-dated commercial terms: price-list reference, credit limit, payment terms, discrepancy window, order lock window, allocation rule	Any processing of money; interest posting (A-12 deferred)
Locations (warehouse, shop, office) with owner, code, language, trading hours, size band, connectivity gate, primary till	Shelf/back-store sub-locations (E-02); organisational hierarchy beyond entity → location
Till positions and devices; assignment; suspension; enrolment hand-off to the sync gateway and the management agent	Device management itself (doc 31), sync (doc 32)
Users, credentials policy, PIN policy, sessions and idle locks, deactivation	Self-service password reset (phase 2); single sign-on with external providers
Permission catalogue, role builder with templates and parameters, scoped assignment, segregation-of-duties pairs, guardrails	Business thresholds themselves (owned by M5/M4 as role parameters they define)
External time-boxed grants for regulator and auditors; Federation view-only class	Regulator return formats (none exist, M-10)
Bulk registration for the onboarding playbook (entities, locations, users, devices) with validation reports	Data cleansing of catalogue (M2, T-03)

2. Requirements coverage

Identifier	Requirement (abridged)	Design section
FR-PTY-010	Entities as legal parties with tier; shop is a location of its entity (ADR-02)	3.1, 3.3
FR-PTY-020	Trading relationships between entities with their own price list and terms (ADR-04)	3.2, 4.3, 6.2
FR-PTY-030	Locations with owner entity, type, language, trading hours	3.3
FR-PTY-040	Till positions per shop; series follow the position (ADR-11)	3.4, 6.6
FR-PTY-050	Devices enrolled, assigned to positions, suspended, retired	3.4, 4.5, 6.6
FR-PTY-060	Credit limit per relationship, informative; changes permission-gated and audited (A-10, A-11, ADR-12)	3.2, 6.4
FR-PTY-070	Entity onboarding status and bulk registration	4.1, 6.1
FR-SEC-010	Dynamic RBAC: permission catalogue, role builder, default templates (N-01, N-02)	3.5, 5.1, 6.3
FR-SEC-020	Till PIN authentication; back-office password; TOTP for financial and administrative functions (N-03, N-04)	3.6, 9.2
FR-SEC-030	Session and idle-lock policy per surface (N-09)	7, 9.2
FR-SEC-040	Deactivate, never delete; history retained (N-08)	4.4
FR-SEC-050	Immutable audit of role, user, credential, grant and device changes (N-05)	5.3 (events), kernel audit
FR-SEC-060	Segregation of duties: requester ≠ approver and configurable pairs (N-07)	3.5, 6.3
FR-SEC-070	No privilege escalation; last-administrator guardrail	3.5
FR-GOV-010	Federation sees everything view-only (M-02 to M-04, T-02)	3.7
FR-GOV-020	Counterparty sees only trading-facing data; shops see nothing of siblings (M-05)	3.7
FR-GOV-030	Each entity self-administers; MPCs administers its shops; no Federation downward admin (M-08, ADR-18)	3.5, 4.4, 6.3
FR-GOV-040	Regulator and auditor time-boxed read-only access, every grant audited (M-10, J-09)	3.7, 4.6, 6.5
FR-GOV-050	Societies cannot leave; data retained indefinitely (M-07)	4.1
FR-GOV-060	Data-governance agreement and responsible officer per entity (M-09, M-11, ADR-26)	3.1 (responsible officer), 11

Identifier	Requirement (abridged)	Design section
NFR-SEC-010/030	ASVS L2; NIST credential rules; MFA	9.2
NFR-CMP-010	PDPA: minimal personal data on users	9.3
NFR-AUD-030	Audit event catalogue coverage for governance events	5.3

3. Domain model

Tables are those of doc 18 part A; this section states the aggregates, their invariants and the two change requests raised against doc 18. Owner of every table here: schema party (entities, relationships, locations, positions, devices) and schema security (users, permissions, roles, assignments, grants, pairs).

3.1 Entity

Aspect	Design
Aggregate	entity with responsible officer and financial-year settings. Root of ownership for everything else.
Invariants	Exactly one FEDERATION row. entity_code immutable and unique. Status never reaches a terminal 'left' state (FR-GOV-050). costing_policy is WEIGHTED_AVERAGE in phase 1.
Change request CR-21-1	Add responsible_officer_user_id (nullable until the officer is appointed) and data_governance_signed_on (date, nullable) to entity for FR-GOV-060; both audited on change.

3.2 Trading relationship

Aspect	Design
Aggregate	entity_relationship (effective-dated) with its terms; the anchor for M3 price lists and M4 exposure.
Invariants	seller ≠ buyer; one ACTIVE row per (seller, buyer) per date (exclusion constraint); a relationship's price_list_id must belong to the seller (validated against M3's query at activation); credit-limit change requires bil.credit_limit.change with MFA and a reason.
Tier rule	Phase 1 permits Federation→distributor and distributor→MPCS. Federation→MPCS is not blocked by schema, only by a configuration flag allow_direct_federation_mpcs (default off), so the change later is data (doc 17 M1 note).
Intra-entity movements	Never a relationship. A transfer between two shops of one MPCS is an M5 document; the schema check is simply seller ≠ buyer.

3.3 Location

Aspect	Design
Aggregate	location with primary_till_position_id.
Invariants	owner_entity_id immutable (A-I2). Only WAREHOUSE and SHOP may hold stock (enforced by M5's query refusing OFFICE). A SHOP needs a primary till before it can issue location-series documents (A-I7). connectivity_spec_met must be true for status ACTIVE.
Language and hours	Language drives the till, receipts and back-office sessions opened at that location (doc 19 section 5.1). trading_hours drives update windows (doc 31) and the day-open that activates apply_from prices (doc 32).

3.4 Till position and device

Aspect	Design
Aggregates	till_position (logical lane) and device (hardware).
Invariants	A position belongs to exactly one SHOP and its position_no is unique there; positions retire, never delete. A device holds at most one position; the assignment change is an audited event with old and new (A-I5). A device below the version floor after grace cannot be assigned to a new position.
Hand-offs	Enrolment creates the device row from the management agent's staging record (doc 31); position assignment triggers series creation through the kernel numbering service (doc 32 section 8); suspension publishes device.suspended.v1 which the sync gateway turns into a signed revoke.

3.5 Users, roles, permissions

Aspect	Design
Aggregates	app_user; permission (catalogue, federation-owned, versioned as a whole); role with role_permission carrying limits jsonb; user_role with scope columns; sod_pair (new, CR-21-2).
Invariants	A user has one home entity. A role is owned by one entity or is a federation template. Scope of an assignment must lie within the grantor's own scope. No user may grant a permission they do not hold. An entity always retains at least one holder of gov.user.manage. FEDERATION-scope permissions appear only in federation-owned roles. An INSTANCE-mode segregation pair is enforced at the acting module (requester ≠ approver on the document); a ROLE-mode pair is enforced here at role authoring.

Aspect	Design
Change request CR-21-2	Add <code>sod_pair</code> (<code>permission_a</code> , <code>permission_b</code> , mode <code>INSTANCE ROLE</code> , <code>owner_entity_id</code> nullable for federation defaults) to doc 18. Defaults: <code>writeoff.request/approve</code> , <code>writeoff.request/witness</code> , <code>adjust.request/approve</code> , <code>user.manage/audit.review(self)</code> at <code>INSTANCE</code> ; <code>role.manage</code> guardrails are code.
Template drift	Role rows cloned from a template keep <code>template_role_id</code> and <code>template_version_seen</code> ; a newer template version shows a marker and a diff; no forced propagation (doc 19 DR-4).

3.6 Credentials

Aspect	Design
Back office	Password hash and TOTP secret live in the identity provider, not in M1's tables; M1 stores only the provider's subject id and status. Credential resets are commands relayed to the provider through the wrapped admin API (doc 19 section 2.2).
Till	<code>pin_hash</code> (Argon2id) stored in M1 for distribution in the operator snapshot; PIN policy from the configuration register: length 4–6, lockout after 5 attempts for 15 minutes, no reuse of last 3 (doc 19 DR-5).

3.7 Visibility policy as data

The four row-level security classes are fixed in code (doc 18 section 3.7); M1 supplies the facts they key on: the caller's scope entity and location, the caller's policy class from the role class, and the `PARTY` relationship table. M1 also owns the column-masking view definitions for `PARTY` reads of trading documents (cost, margin and internal notes omitted), published as a versioned kernel artefact so M4 can rely on them.

Class	Who	Reads	Writes
OWN	Any user in scope	Rows where <code>owner</code> = scope entity (and location, when the assignment is location-scoped)	Per permission
PARTY	Seller or buyer of a document	Trading-facing columns of documents where the scope entity is a party	Their own side's commands only
FEDERATION_VIEW	Federation roles marked view-all	Everything	None (except Federation-owned reference data through OWN)
EXTERNAL_TIMEBOXED	Regulator, auditors	Granted scope, read-only, until <code>valid_until</code>	None

4. State machines

M1 owns no trading documents; its lifecycles are status machines on reference records. They are small but every transition is permission-gated, audited and has a reason rule where doc 03 conventions require one.

4.1 Entity

From	To	Trigger / command	Guard	Permission	Audit
–	ONBOARDING	RegisterEntity	Unique code; tier valid	gov.entity.register (FEDERATION)	entity.registered
ONBOARDING	ACTIVE	ActivateEntity	Responsible officer named; at least one admin user; VAT registration present	gov.entity.activate (FEDERATION)	entity.activated
ACTIVE	SUSPENDED	SuspendEntity	Reason code mandatory	gov.entity.suspend (FEDERATION, MFA)	entity.suspended
SUSPENDED	ACTIVE	ReinstateEntity	Reason code	gov.entity.suspend (FEDERATION, MFA)	entity.reinstated

There is no CLOSED state (FR-GOV-050). Suspension blocks new documents by the entity; existing documents and history remain readable.

4.2 Trading relationship

From	To	Trigger / command	Guard	Permission	Audit
–	DRAFT	OpenTradingRelationship	seller ≠ buyer; tier rule or flag; no overlapping ACTIVE row	prt.relationship.open (seller)	relationship.opened
DRAFT	ACTIVE	ActivateRelationship	Price list belongs to seller and is published; terms complete	prt.relationship.activate (seller)	relationship.activated
ACTIVE	ACTIVE (new row)	AmendRelationshipTerms	Effective-dated: old row closed, new row opened; credit-limit change needs MFA and reason	prt.relationship.amend; bil.creditlimit.change for the limit	relationship.amended; credit_limit.changed
ACTIVE	SUSPENDED	SuspendRelationship	Reason; open documents unaffected	prt.relationship.suspend (seller, MFA)	relationship.suspended

4.3 Location

From	To	Trigger / command	Guard	Permission	Audit
–	PLANNED	RegisterLocation	Owner ACTIVE or ONBOARDING; code unique in owner	prt.location.register (owner admin)	location.registered
PLANNED	ONBOARDING	StartOnboarding	Device staged	prt.location.onboard	location.onboarding
ONBOARDING	ACTIVE	ActivateLocation	Connectivity gate met; primary till set for SHOP; operators assigned	prt.location.activate	location.activated
ACTIVE	DORMANT	MarkDormant	No open till session; reason	prt.location.dormant	location.dormant
DORMANT	ACTIVE	Reactivate	Connectivity gate met	prt.location.activate	location.activated

4.4 User

From	To	Trigger / command	Guard	Permission	Audit
–	PENDING	CreateUser	Home entity in grantor's scope; unique username in federation	gov.user.manage (MFA)	user.created
PENDING	ACTIVE	First credential set (provider callback) or PIN issued	–	system	user.activated
ACTIVE	LOCKED	Lockout (system)	Failed attempts over policy	system	account.locked_out

From	To	Trigger / command	Guard	Permission	Audit
LOCKED	ACTIVE	Unlock / ResetCredential	Administrator in scope	gov.user.manage (MFA)	user.credential_reset
ACTIVE	DEACTIVATED	DeactivateUser	Not the last user.manage holder; reason	gov.user.manage (MFA)	user.deactivated

DEACTIVATED is terminal; a returning member of staff receives a new user record linked by succeeds_user_id so history stays attributable to the original.

4.5 Device

From	To	Trigger / command	Guard	Permission	Audit
–	ENROLLED	EnrolDevice (from staging)	Hardware serial unique; management agent attested	sys.device.enrol (MFA)	device.enrolled
ENROLLED	ACTIVE	AssignDeviceToPosition	Position free; device above floor	sys.device.enrol	device.assigned
ACTIVE	ACTIVE	AssignDeviceToPosition (re-assign)	Old device's outbox drained or loss recorded (doc 32 section 8)	sys.device.enrol (MFA)	device.position_changed; series_holder_changed
ACTIVE	SUSPENDED	SuspendDevice	Reason (lost, stolen, faulty)	sys.device.suspend (MFA)	device.suspended
SUSPENDED	ACTIVE	ReinstateDevice	Physical recovery confirmed	sys.device.suspend (MFA)	device.reinstated
ANY	RETIRED	RetireDevice	Unassigned; wiped	sys.device.suspend (MFA)	device.retired

4.6 External grant

From	To	Trigger / command	Guard	Permission	Audit
–	ACTIVE	GrantExternalView	Grantee user of class EXTERNAL; scope; valid_until ≤ 12 months; reason	gov.external.grant (FEDERATION, MFA)	external_grant.issued
ACTIVE	EXPIRED	Clock (system)	valid_until passed	system	external_grant.expired
ACTIVE	REVOKED	RevokeExternalView	Reason	gov.external.grant (MFA)	external_grant.revoked

5. Contract

5.1 Commands

Command	Input (essentials)	Permission	Effect	Events
RegisterEntity	code, type, names, registration, VAT no, district, defaults	gov.entity.register (F)	Entity ONBOARDING	entity.registered.v1
ActivateEntity / SuspendEntity / ReinstateEntity	entity, reason	gov.entity.activate / .suspend (F, MFA)	Status	entity.activated/suspended/reinstated.v1
AppointResponsibleOfficer	entity, user, signed date	gov.entity.register (F) or gov.user.manage (own)	CR-21-1 fields	entity.updated.v1
OpenTradingRelationship	seller, buyer, price list, terms, effective_from	prt.relationship.open	DRAFT row	relationship.opened.v1
ActivateRelationship	relationship	prt.relationship.activate	ACTIVE; M3 query validates list	relationship.activate.v1
AmendRelationshipTerms	relationship, new terms, effective_from, reason	prt.relationship.amend; bil.creditlimit.change (MFA) when limit changes	Close old row at effective_from – 1, open new	relationship.amended.v1; credit_limit.changed.v1
SuspendRelationship	relationship, reason	prt.relationship.suspend (MFA)	Status	relationship.suspended.v1
RegisterLocation / ActivateLocation / MarkDormant	owner, code, type, names, address, language, hours, size band	prt.location.* (owner admin)	Status per 4.3	location.*.v1
SetPrimaryTill	location, position, reason	prt.location.primary (owner admin, MFA)	Updates flag; kernel moves location series	location.primary_changed.v1; series.holder_changed.v1
RegisterTillPosition / RetireTillPosition	location, position_no	prt.position.manage	Position row; kernel creates or closes series	till_position.registered/retired.v1
EnrolDevice	serial, kind, owner (from staging)	sys.device.enrol (MFA)	Device ENROLLED	device.enrolled.v1
AssignDeviceToPosition	device, position, reason	sys.device.enrol (MFA on re-assign)	Assignment; counter transfer via kernel	device.assigned/position_changed.v1
SuspendDevice / ReinstateDevice / RetireDevice	device, reason	sys.device.suspend (MFA)	Status; gateway revoke	device.suspended/reinstated/retired.v1
CreateUser	home entity, username, display name, language, kind (BACK_OFFICE TILL BOTH)	gov.user.manage (MFA)	PENDING; provider account created via wrapped admin API	user.created.v1
ResetCredential	user, kind	gov.user.manage (MFA)	Temporary password via provider, or new PIN hash	user.credential_reset.v1
DeactivateUser	user, reason	gov.user.manage (MFA)	DEACTIVATED; provider disabled; refresh tokens revoked	user.deactivated.v1
CreateRole / AmendRole / RetireRole	owner entity, name, permissions with limits, template ref	gov.role.manage (MFA)	Role rows; guardrails	role.changed.v1
AssignRole / RevokeRole	user, role, scope entity, scope location	gov.role.manage (MFA)	user_role row	role.assigned/revoked.v1
SetSodPairMode	pair, mode	gov.role.manage (MFA)	sod_pair row	sod_pair.changed.v1
GrantExternalView / RevokeExternalView	grantee, scope, valid_until, reason	gov.external.grant (F, MFA)	Grant row	external_grant.issued/revoked.v1
BulkRegister	file (entities, locations, positions, users, devices)	gov.entity.register (F) or gov.user.manage (own)	Validated staging; commit per row; validation report	the per-row events

F = FEDERATION scope only. All commands are idempotent by key (doc 14 section 1.3).

5.2 Queries

Query	Returns	Scope rule
ResolveScope(user)	Home entity, scopes (entity, location) list, policy class	Kernel only
ResolvePermissions(user, rv)	Effective permission set with limits, per scope	Kernel and till snapshot builder
LookupRelationship(seller, buyer, date)	Active relationship with terms	Parties and Federation view
ListRelationships(entity, role)	As seller or buyer	OWN
GetEntity / ListEntities(filter)	Entity facts	OWN, PARTY (names only), FEDERATION_VIEW
ListLocations(entity) / GetLocation	Locations with primary till, hours, language	OWN; PARTY sees ship-to names for drops
ListOperators(location)	Users assigned at the location with PIN hashes and offline permissions	Kernel snapshot builder only
ListDevices(entity)	Devices, positions, versions, last seen (last seen from M6 health)	OWN; FEDERATION_VIEW
ListRoles(entity) / GetRoleDiff(role)	Roles and template diffs	OWN
ListExternalGrants()	Active and expired grants	FEDERATION

5.3 Events published

Event	Payload (beyond envelope)	Consumers
entity.registered/activated/suspended/reinstated/updated.v1	entity id, code, type, status	Numbering (entity series), M3, M4, M8
relationship.opened/activated/amended/suspended.v1; credit_limit.changed.v1	relationship id, parties, effective range, terms hash; limit old/new	M3 (price list binding), M4 (exposure), M8
location.registered/onboarding/activated/dormant/primary_changed.v1	location id, owner, type, language, hours, primary position	Numbering, sync gateway (snapshot scope), doc 31 windows, M5, M8
till_position.registered/retired.v1; series.holder_changed.v1 (kernel)	position id, location, number	Numbering, sync gateway
device.enrolled/assigned/position_changed/suspended/reinstated/retired.v1	device id, serial, position, reason	Sync gateway (revoke, counter transfer), doc 31 fleet
user.created/activated/credential_reset/deactivated.v1; account.locked_out.v1; authentication.failed.v1	user id, home entity, kind (no credential material)	Sync gateway (operator snapshot), audit exception queues
role.changed/assigned/revoked.v1; sod_pair.changed.v1	role id, owner, version, permission diff	Sync gateway (permission snapshot), audit
external_grant.issued/expired/revoked.v1	grant id, grantee, scope, valid_until	Audit, M8

6. Key flows

Flow	Sequence	Failure paths
6.1 Onboarding an MPCS with three shops (playbook, about 40 shops a week)	Federation runs BulkRegister from the onboarding sheet → entity ONBOARDING, locations PLANNED, positions, staged devices → MPCS administrator user created with a temporary credential → MPCS admin creates shop users and assigns roles from templates (shop-in-charge for single-staff shops) → devices enrolled at Federation staging and assigned → connectivity gate recorded per site → ActivateLocation per shop → ActivateEntity once the responsible officer is named	Validation report lists rejected rows (duplicate codes, missing VAT no, invalid district); nothing partial commits per row; a shop without a primary till cannot activate; an entity without an officer stays ONBOARDING but its shops may trade under the distributor relationship only once the entity is ACTIVE
6.2 Opening a distributor→MPCS relationship	Distributor sales opens DRAFT with terms and a negotiated price list (M3) → ActivateRelationship validates the list belongs to the seller and is published → M4 may now accept orders from the buyer	Overlapping ACTIVE row rejected by the exclusion constraint; unpublished list rejected with the M3 reason
6.3 Building a role for a shop	MPCS admin clones the shop-in-charge template → adds prt.count.record limits → guardrail checks: no permission beyond grantor's own, no FEDERATION permission, no ROLE-mode pair conflict → AssignRole scoped to the shop location	A conflict names the pair and the mode; assignment outside the grantor's scope is refused
6.4 Raising a credit limit	Seller accounts opens AmendRelationshipTerms with the new limit and reason → step-up TOTP → new effective-dated row → credit_limit.changed.v1 → M4 recomputes exposure warning	Missing reason or stale MFA rejected; the old row is never edited
6.5 Regulator access for an inspection	Federation admin creates an EXTERNAL user → GrantExternalView with scope and valid_until → every login and export audited → expiry by clock	Grant beyond 12 months refused; revocation immediate; export volume visible on the exception queue
6.6 Replacing a faulty till	Administrator suspends the old device (stolen or faulty) → if its outbox was not drained, records the loss (doc 32 sequence reset) → new device enrolled at staging and shipped → AssignDeviceToPosition triggers counter transfer from the highest applied number → old device retired	A drained-check failure blocks assignment until the loss is recorded; series gap documented, never hidden

7. Configuration

Item	Default	Scope	Who may change
allow_direct_federation_mpcs	off	Federation	gov.entity.register (F)
pin_length_min / max	4 / 6	Entity	gov.user.manage
pin_lockout_attempts / minutes	5 / 15	Entity	gov.user.manage
idle_lock_till_sales / shop_mode / back_office / admin (minutes)	2 / 3 / 10 / 5	Entity (till values per location override)	gov.user.manage
mfa_freshness_hours	12; per-action list: credit limit, control price, external grant	Federation default; entity may shorten	gov.role.manage
external_grant_max_months	12	Federation	gov.external.grant
sod_pair modes	INSTANCE for the four defaults	Entity	gov.role.manage
template_drift_notification	notify and diff	Federation	—
location.trading_hours, language, size_band, connectivity_spec_met	per site	Location	prt.location.*

8. Screens as paper

Screen	Paper mirrored	Surface	Roles	Notes
Society register	Register of societies	W	Federation admin	List, detail, status actions; onboarding checklist tile (officer, admin user, VAT, shops active)
Agreement sheet (relationship)	Trading agreement	W	Seller admin, accounts	Terms as a form; effective-dated history as a timeline; limit change opens ReasonCapture and step-up
Shop list and shop card	Shop register	W	Entity admin	Hours, language, size band, connectivity gate, primary till selector (audited), tills and devices
Staff register	Staff book	W	Entity or MPCS admin	Create, deactivate, reset; roles shown as duties in the shop language; scope shown as shop names

Screen	Paper mirrored	Surface	Roles	Notes
Duties list (role builder)	Duty roster	W	Entity admin	Permissions grouped by paper document (receipt, GRN, count, write-off, price list); limits inline; template diff banner
Asset register (devices)	Asset register	W	Entity admin	Serial, position, version, last seen, suspend and reassign with reason
Access letter (external grants)	Access letter	W	Federation admin	Grantee, scope, until; audit trail of use
Bulk registration	Onboarding sheet	W	Federation, entity admin	Upload, validation report with row-level reasons, commit
Exception queue	Exceptions book	W	Named reviewers	Shared component; governance events (lockouts, grants, role changes)

No M1 screen exists on the till; the till consumes the operator and permission snapshots only.

9. Non-functional design

Aspect	Design
9.1 Capacity	876 entities, about 10,000 locations, 15,000 positions and devices, tens of thousands of users; all small tables. ResolvePermissions is cached per instance keyed by catalogue version; ListOperators runs per snapshot build, not per request.
9.2 Security	All mutating commands MFA-gated except location and position registration; provider admin API wrapped so no Federation role reaches it; PIN hashes Argon2id with per-user salt; credential material never in events or audit; lockouts audited; ASVS L2 controls mapped in doc 34.
9.3 Privacy	User records hold display name, username, language and kind; no personal identifiers beyond what the identity provider needs; responsible officer is a user reference. PDPA data map in doc 34.
9.4 Availability	M1 reads are on every request path; the permission and scope caches make M1 unavailability survivable for reads; writes are rare.
9.5 Auditability	Every command in 5.1 raises a catalogue event; role changes carry before/after permission sets; grants carry scope and validity.

10. Test strategy

Layer	What is tested
Unit	Guardrails: no escalation, last-administrator, FEDERATION-only permissions, tier rule and flag, seller ≠ buyer, effective-date closure arithmetic
Contract	Every command and query in section 5 against the published schema; idempotency by key; MFA step-up challenge on the gated commands
State machine	Every transition in section 4 with its guard; illegal transitions rejected with the reason code
Property-based	Random sequences of role edits never produce a role holding a ROLE-mode pair; random assignment scopes never exceed the grantor's; exclusion constraint holds under random effective ranges
Row-level security matrix	Principal × table × operation across the four classes: a distributor user cannot read MPCs internal rows; a shop-scoped user cannot read a sibling shop; Federation writes nothing; an expired grant reads nothing
Scenario	Flows 6.1–6.6 end to end, including the counter-transfer hand-off to the kernel and the revoke hand-off to the sync gateway
Performance	ResolvePermissions under 2 ms from cache; BulkRegister of 40 shops with 120 users and 60 devices under 60 s with a full validation report

11. Decision requests

#	Point	Proposed	Decider
DR-1	Responsible officer and data-governance date on entity (CR-21-1)	Adopt; nullable until appointed; activation requires the officer	Architecture review; Federation legal
DR-2	Segregation pairs as data (CR-21-2)	Adopt with the four INSTANCE defaults	Architecture review
DR-3	Federation→MPCS direct trading flag	Present, default off	Federation admins
DR-4	External grant maximum duration	12 months	Federation, Department of Co-operative Development liaison
DR-5	Returning staff: new user record with succession link vs reactivation	New record with succeeds_user_id	Federation admins
DR-6	Entity activation prerequisites	Officer named, admin user, VAT registration	Federation admins

12. Traceability

Section	Identifiers
3.1–3.4	FR-PTY-010 to 070; ADR-02, ADR-04, ADR-11, ADR-12; A-06, A-07, A-09 to A-11, D-06, C-04, C-08; M-07
3.5–3.7	FR-SEC-010, 060, 070; FR-GOV-010 to 040; N-01, N-02, N-07; M-02 to M-05, M-08, M-10; ADR-18, ADR-26; doc 18 3.7; doc 19 3
4	Doc 03 conventions; N-08; FR-SEC-040; FR-GOV-050; doc 32 section 8
5	Doc 20 M1 contract; doc 19 6.1; doc 14 1.3

Section	Identifiers
6	S-07 (40 shops a week); doc 13 scenarios where relationships and roles appear
7	Doc 07 section 4 configuration register; N-03, N-09; doc 19 DR-4, DR-5, DR-7
8	Doc 30 section 5.1; P-08
9–10	NFR-SEC, NFR-CMP, NFR-AUD; doc 34, doc 39

End of document. Draft for design review; the contract in section 5 freezes when at least one dependant (M2) acknowledges it, targeted end of Stage 2 week 2.